Arista Access Node WFD-ACC-44 Routing Protocol Flap SOP (Major)

Node: WFD-ACC-44 | **Vendor:** Arista | **Severity:** Major | **Alarm Code:** ALARM_2686

Description

This Standard Operating Procedure (SOP) is designed to mitigate a critical routing protocol flap failure on the Arista Access Node WFD-ACC-44 at the Whitefield POP site. The primary failure domain in this scenario involves BGP route churn, OSPF adjacency loss, and ISIS LSP flooding. These protocol flaps can occur due to a variety of reasons including routing loop detection, neighbor reset, or routing table instability. On Arista equipment, this can be particularly problematic as it can lead to an increase in CPU utilization, memory usage, and potentially cause traffic blackholing. The root cause of the issue could be attributed to misconfigured routing policies, incorrect interface configuration, or even hardware-related issues. In this SOP, we will guide the engineer through a series of diagnostic and resolution steps to isolate and rectify the issue, ensuring the network remains stable and traffic-forwarding capabilities are preserved.

Pre-requisite Checks

- Requirement: Verify the network team has dispatched remote hands to the Whitefield POP site to facilitate physical access to the WFD-ACC-44 node., Status: TBD

- Requirement: Confirm that redundant links are operational and not affected by the failure., Status: TBD

Potential Root Causes

Cause	Probability
Misconfigured routing policies causing routing loop detection.	High
Neighbor reset or routing table instability due to hardware or software issues.	Medium
Incorrect interface configuration or suboptimal switch configuration.	Low

Diagnostic Commands & Expected Outputs

```
Command: show bgp summary, Expected Output: BGP table version is 12
BGP table version 12 Status codes: s-suppressed, d-dampened, h-hold,
*-valid, >-best, i-internal, r-RIB-failure, S-stale Origin codes: i -
IGP, e - EGP, ? - incomplete Network Next Hop Metric LocPrf Weight
Path *> 10.0.0.0/24 10.10.10.10 100 100 32768 i *> 10.1.1.1/32
10.10.10.10 100 100 32768 i, Description: This command displays a
summary of the BGP routing table, including the next hop, metric, and
path. It will help identify any route churn or instability.

Command: show ospf neighbor, Expected Output: Neighbor ID Pri State
Dead Time Address Interface 10.10.10.10 128 FULL/ - 00:00:32
10.10.10.10 Eth1/1 , Description: This command displays a list of OSPF
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```
neighbors, their state, and the interface they are connected to. It
will help determine if any OSPF adjacency loss has occurred.

Command: show isis database, Expected Output: ISIS LSP Database
(Sequence/LSU ID) 1000/0x12345678
5.1000.0001.0000.0000.0000.0000.0000.00 , Description: This command
displays the ISIS LSP database, which will help identify any LSP
flooding or instability.
```

Resolution Steps

Step 1: Step: Verify and correct routing policies, Command: show running-config | grep -E 'route-map|policy', Description: Review the running configuration to identify any misconfigured routing policies. If found, correct the policies and reapply the configuration changes.

Step 2: Step: Verify and correct neighbor settings, Command: show bgp neighbors 10.10.10.10, Description: Review the BGP neighbor settings to ensure they are correct. If found to be incorrect, correct the settings and reapply the configuration changes.

Step 3: Step: Verify and correct interface settings, Command: show interface Eth1/1, Description: Review the interface settings to ensure they are correct. If found to be incorrect, correct the settings and reapply the configuration changes.

Post-Check Verification

```
Command: show bgp summary, Expected Output: BGP table version is 12
BGP table version 12 Status codes: s-suppressed, d-dampened, h-hold,
*-valid, >-best, i-internal, r-RIB-failure, S-stale Origin codes: i -
IGP, e - EGP, ? - incomplete Network Next Hop Metric LocPrf Weight
Path *> 10.0.0.0/24 10.10.10.10 100 100 32768 i *> 10.1.1.1/32
10.10.10.10 100 100 32768 i, Description: Verify that the BGP routing
table is stable and no route churn is occurring.

Command: show ospf neighbor, Expected Output: Neighbor ID Pri State
Dead Time Address Interface 10.10.10.10 128 FULL/ - 00:00:32
10.10.10.10 Eth1/1 , Description: Verify that the OSPF neighbors are
stable and no adjacency loss has occurred.

Command: show isis database, Expected Output: ISIS LSP Database
(Sequence/LSU ID) 1000/0x12345678
5.1000.0001.0000.0000.0000.0000.0000.00 , Description: Verify that the
ISIS LSP database is stable and no LSP flooding is occurring.
```

Roll-back Procedures

- Step: Restore previous configuration, Command: copy running-config startup-config, Description: Restore the previous configuration from the startup configuration.

- Step: Verify network stability, Command: show bgp summary, Description: Verify that the BGP routing table is stable and no route churn is occurring.